

Quantum Machine Learning for Pneumonia Detection in Chest X-Ray Imaging: an Illustrative Case Study

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Abstract

Deep convolutional neural networks (CNNs) have become the de facto standard for automated pneumonia screening from chest X-ray (CXR) images, routinely reporting diagnostic accuracies above 85%. This performance, however, is obtained at the cost of classification heads with hundreds of thousands of trainable parameters, motivating interest in parameter-efficient alternatives. Variational quantum circuits (VQCs), operating within the constraints of Noisy Intermediate-Scale Quantum (NISQ) hardware, have been proposed as compact decision layers that can be appended to classical feature extractors. Beyond the pure VQC paradigm, quantum convolutional neural networks (QCNNs) and quantum kernel methods have been explored as complementary architectural routes. This paper reviews the computational and algorithmic foundations of such hybrid quantum–classical architectures for CXR-based pneumonia detection, covering classical CNN baselines, quantum data-encoding strategies (angle, amplitude, IQP feature maps, and data re-uploading), variational and convolutional ansatz design, quantum kernel construction, and the barren-plateau phenomenon that limits the trainability of deep parametrized circuits together with its principal mitigation strategies. As a concrete illustration, we describe and analyze a hybrid pipeline that couples a frozen ConvNeXt-Tiny feature extractor with domain-adversarial representation learning and nonlinear autoencoder dimensionality reduction (768→64) feeding a 6-qubit, 62-parameter data re-uploading VQC classifier, evaluated on a publicly available pediatric CXR dataset under an explicit train/test class-prevalence shift. On a patient-disjoint held-out test set of 624 images, the hybrid model attains a test balanced accuracy of 0.747 and a test AUC-ROC of 0.806, against 0.729 and 0.848 for a classical multilayer perceptron trained on identical features—a difference that is not statistically significant (McNemar’s exact test, $p = 0.71$)—while using approximately $34\times$ fewer trainable parameters in its decision layer (62 versus 2,113). We situate this case study within a broader survey of related hybrid approaches for respiratory and cardiac CXR analysis reported in the literature, and ground a discussion of the algorithmic trade-offs, evaluation methodology, and open problems—including state-preparation overhead, barren plateaus, dataset shift, and the absence, at present circuit scales, of a demonstrated computational quantum advantage. Finally, we report the execution of the trained circuit on a physical IBM Quantum processor and quantify the simulator-to-hardware gap. On a class-balanced subset, hardware noise compresses the span of the predicted pneumonia probability by 59% relative to the noise-free statevector simulator ($0.439 \rightarrow 0.182$); we attribute this primarily to the 62 two-qubit gates required for amplitude-encoding state preparation rather than to the 18 gates of the learned ansatz, which accounts for under a quarter of the circuit’s two-qubit budget. Zero-noise extrapolation restores the distributional span but not the sample-level agreement with the simulator, indicating that error mitigation at this circuit scale mitigates variance at the cost of additional bias.

Keywords: quantum machine learning; variational quantum circuits; hybrid quantum–classical neural networks; quantum convolutional neural networks; quantum kernel methods; pneumonia detection; chest X-ray classification; NISQ; barren plateaus; medical image analysis

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1 Introduction

Pneumonia remains one of the leading causes of morbidity and mortality worldwide, with an estimated global burden in the range of millions of cases annually and disproportionate impact on children under five and immunocompromised adults [3, 12]. Chest radiography (CXR) is the most widely used first-line imaging modality for pneumonia screening because it is inexpensive, fast, and broadly available compared to computed tomography [12]. Manual interpretation of CXR images is subject to inter-observer variability, reader fatigue, and uneven access to trained radiologists, which has motivated three decades of research into computer-aided diagnosis and, more recently, deep-learning-based automated screening [23, 16].

CNNs, and in particular deep residual architectures such as ResNet, have demonstrated diagnostic performance on CXR pneumonia classification that is competitive with expert radiologists, typically exceeding 85% classification accuracy [10, 16, 34]. This performance, however, comes at a substantial computational cost: fully connected classification heads appended to CNN backbones commonly require on the order of 10^5 – 10^6 trainable parameters. The associated memory and energy costs are non-trivial for edge or point-of-care deployment scenarios and motivate the search for parameter-efficient decision layers.

In parallel, the maturation of NISQ devices has generated interest in variational quantum algorithms (VQAs) as parameter-efficient function approximators [31, 28]. A VQC parametrizes a family of unitary transformations $U(\theta)$ and, when combined with classical optimization of θ , can implement expressive decision boundaries using a number of trainable parameters that grows only linearly with circuit depth and qubit count [21, 2]. This has prompted a growing body of work on *hybrid quantum-classical* (HQC) architectures, in which a classical CNN performs high-dimensional feature extraction and a quantum component performs low-dimensional classification, confining the quantum computation to regimes tractable on current simulators or hardware.

A number of recent studies have applied HQC architectures specifically to thoracic imaging tasks. Kulkarni et al. [14] demonstrated that replacing a classical CNN layer with a VQC yields statistically significant improvements for pneumonia detection on CXR images. Rao et al. [30] reported a 98.1% test accuracy on a COVID-19 Radiography Dataset using a hybrid CNN–VQC pipeline deployed partly on an IBM Q simulator, while Decoodt et al. [6] achieved an AUC of up to 0.93 for cardiomegaly detection using a DenseNet-121 backbone augmented with a variational quantum layer. At the same time, attention has shifted beyond the “quantum classifier head” paradigm: quantum convolutional neural networks (QCNNs) designed to mimic spatial locality in quantum circuits [5, 38], and quantum kernel methods that compute high-dimensional inner products via quantum interference [9, 8], represent architecturally distinct alternatives that may better exploit the inductive biases of image data.

Despite this activity, empirical evidence quantifying the accuracy–efficiency trade-off of HQC pipelines on realistic medical imaging tasks—particularly under distribution shift between training and deployment data—remains limited. Most published studies report results on a single static train/test split, on noise-free simulators, and without explicit discussion of the state-preparation and trainability bottlenecks intrinsic to variational quantum computing [35].

1.1 Contributions and Scope

This paper has three purposes. First, it provides a structured, computationally oriented review of the building blocks required to construct hybrid quantum–classical pneumonia classifiers: classical CNN backbones, dimensionality reduction, quantum data-encoding strategies, variational and convolutional ansatz design, quantum kernel construction, and the barren-plateau problem including its principal mitigation strategies. Second, it presents a detailed case study—a ConvNeXt-Tiny/autoencoder/VQC pipeline evaluated on a public pediatric CXR dataset—that grounds the algorithmic discussion with explicit numbers and illustrates the current gap

between classical and quantum-assisted approaches. Third, it closes the simulator-to-hardware loop that most simulation-only studies leave open, by executing the trained circuit on a physical IBM Quantum processor and reporting the resulting degradation quantitatively, together with a gate-level attribution of its cause.

The clinical and radiological background is kept intentionally brief; the emphasis throughout is on encoding schemes, circuit design, trainability, computational complexity, and evaluation methodology, consistent with a computer-science- and quantum-technology-oriented readership. We deliberately avoid overstating the clinical readiness of the reviewed methods: at the qubit counts considered here, no computational quantum advantage is established, and the results should be interpreted as a feasibility and parameter-efficiency study rather than as a validated diagnostic tool.

The remainder of the paper is organized as follows. Section 2 introduces the clinical context and the classical and quantum computational preliminaries. Section 3 reviews classical CNN-based approaches to CXR pneumonia detection. Section 4 describes quantum data encoding, variational circuits, QCNs, quantum kernels, and the barren-plateau phenomenon. Section 5 surveys related HQC work for CXR analysis and presents the ConvNeXt-Tiny/autoencoder/VQC case study. Section 6 reports the physical-hardware execution and quantifies the simulator-to-hardware gap. Section 7 discusses experimental design and evaluation metrics. Section 8 enumerates open problems and limitations. Section 9 outlines future research directions, and Section 10 concludes.

2 Background

2.1 Clinical and Imaging Context

Pneumonia is an acute inflammatory condition of the pulmonary parenchyma, most commonly of bacterial or viral etiology, in which the alveoli fill with fluid or exudate, impairing gas exchange [12]. On a CXR, healthy air-filled lung tissue appears radiolucent (dark), whereas pneumonic consolidation, interstitial infiltrates, and air bronchograms appear as localized or diffuse radiopaque (light) regions [12]. Bacterial pneumonia (e.g., *Streptococcus pneumoniae*) typically presents with lobar consolidation, while viral pneumonia tends to produce more diffuse interstitial patterns [12]. This visual heterogeneity, combined with overlapping appearances with other conditions (pulmonary edema, atelectasis, fibrosis), contributes to inter-observer variability in manual reading and motivates automated decision support [23].

2.2 Classical Deep Learning for Medical Image Classification

Modern CXR classifiers are almost universally built on deep CNN backbones pretrained on large natural-image corpora (e.g., ImageNet) and subsequently fine-tuned or used as fixed feature extractors on medical imaging tasks. ResNet architectures address the vanishing-gradient and degradation problems that historically limited the trainability of very deep networks by introducing residual (skip) connections: instead of learning a direct mapping $H(x)$, each residual block learns a residual function $F(x)$ such that

$$H(x) = F(x) + x, \quad (1)$$

which allows gradients to propagate across many layers without attenuation [10]. ResNet-50 uses a bottleneck design in which each residual block consists of a 1×1 convolution (dimensionality reduction), a 3×3 convolution (spatial filtering), and a further 1×1 convolution (dimensionality restoration), substantially reducing parameter count relative to a naive stacking of 3×3 convolutions while preserving representational depth [10]. In a typical CXR pipeline, the network is used *headless*, i.e., without its final classification layer, producing a fixed-length

embedding (commonly of dimension 2048 for ResNet-50) for each input image, which is then passed to a task-specific classifier.

2.3 Elements of Quantum Computation

A qubit is the basic unit of quantum information, represented as a normalized vector in a two-dimensional complex Hilbert space,

$$|\psi\rangle = \alpha|0\rangle + \beta|1\rangle, \quad |\alpha|^2 + |\beta|^2 = 1, \quad (2)$$

where $\alpha, \beta \in \mathbb{C}$ are probability amplitudes and $\{|0\rangle, |1\rangle\}$ is the computational basis [24]. Geometrically, a pure single-qubit state can be represented as a point on the Bloch sphere,

$$|\psi\rangle = \cos\left(\frac{\theta}{2}\right) |0\rangle + e^{i\varphi} \sin\left(\frac{\theta}{2}\right) |1\rangle, \quad (3)$$

and training a variational circuit can be understood, at the single-qubit level, as searching for rotation angles that steer this state toward a measurement outcome associated with the desired class.

Quantum algorithms are composed of unitary *gates*. Single-qubit rotation gates R_x, R_y, R_z (or the three-angle rotation $\text{Rot}(\phi, \theta, \omega)$) rotate the state around the corresponding axis of the Bloch sphere and constitute the trainable degrees of freedom of a variational circuit; the Hadamard gate creates equal superpositions; and the two-qubit controlled-NOT (CNOT) gate is the standard primitive for generating *entanglement*, a distinctively quantum correlation between qubits that cannot be reproduced by any local, classical description of the joint system and that is conjectured to underlie the representational power of quantum models [24].

The *expressibility* of a parametrized circuit is a quantitative measure of how uniformly it can generate states over the Hilbert space, often defined via the deviation of the induced distribution from the Haar measure [37]. Expressibility and entanglement capacity jointly govern the circuit’s ability to represent complex functions and, as detailed in Section 4.5, its susceptibility to trainability issues.

2.4 The NISQ Era and Variational Quantum Algorithms

Contemporary quantum processors operate in what Preskill termed the Noisy Intermediate-Scale Quantum (NISQ) era: devices with tens to a few hundred physical qubits that are not yet fault-tolerant and remain subject to gate errors and decoherence [28]. Variational quantum algorithms are the dominant algorithmic paradigm for extracting practical value from NISQ hardware. A VQA defines a parametrized unitary $U(\theta)$, executes it on a (possibly classically simulated) quantum device, measures an observable, and updates θ using a classical optimizer—a hybrid quantum–classical optimization loop that keeps quantum resource requirements shallow while delegating gradient-based optimization to classical compute [31, 21]. Variational Quantum Classifiers (VQCs) are the supervised-learning instance of this paradigm and form the quantum component of the hybrid architectures discussed in this paper.

3 Classical Approaches to Pneumonia Detection from Chest X-Rays

The dominant classical paradigm for CXR-based pneumonia detection follows a transfer-learning recipe: (i) a CNN backbone pretrained on a large natural-image dataset is used, either fine-tuned end-to-end or frozen as a fixed feature extractor; (ii) a fully connected classification head, typically consisting of one or more dense layers followed by a sigmoid or softmax output, is trained on the target task; and (iii) standard image preprocessing (resizing, normalization with respect

to the pretraining dataset’s channel statistics) is applied to bridge the domain gap between natural and medical images. ResNet, DenseNet, and, more recently, vision-transformer backbones have all been applied to the publicly available pediatric CXR pneumonia dataset originating from the Guangzhou Women and Children’s Medical Center, with reported accuracies generally exceeding 85% and F1 scores in the 0.85–0.92 range [16, 34, 23].

From a purely computational standpoint, the classification head is the component of the pipeline most directly comparable to a quantum decision layer. For a 64-dimensional input (e.g., after dimensionality reduction, see Section 5) and a single hidden layer of width 32, a standard multilayer perceptron (MLP) classification head requires

$$P_{\text{MLP}} = (64 \times 32 + 32) + (32 \times 1 + 1) = 2,080 + 33 = 2,113 \text{ parameters}, \quad (4)$$

illustrating that even a comparatively shallow classifier on a compressed feature space carries a non-trivial parameter budget. This observation is the principal motivation for exploring VQC-based decision layers, whose parameter count scales as $O(nL)$ in the number of qubits n and circuit layers L , rather than with the width of a dense layer (Section 4).

Two further classical design choices are directly relevant to the hybrid architectures discussed below. First, class imbalance—pneumonia typically constitutes 70–75% of samples in public pediatric CXR datasets—motivates the use of weighted loss functions (e.g., weighted binary cross-entropy) rather than unweighted cross-entropy, to avoid a majority-class bias. Second, dimensionality reduction (via PCA, linear discriminant analysis, or univariate feature selection) is frequently required before any quantum processing step, since CNN embeddings (commonly 512–2048-dimensional) exceed the practical qubit budgets of NISQ-era hardware and simulators by orders of magnitude.

4 Quantum Machine Learning Approaches

4.1 Data Encoding Strategies

Before a classical feature vector $x \in \mathbb{R}^N$ can be processed by a quantum circuit, it must be mapped, via a *state-preparation* routine, onto a quantum state $|\psi(x)\rangle$ in an n -qubit Hilbert space. The choice of encoding is consequential: it determines the geometry of the induced kernel, the depth of the state-preparation circuit, and, as recent theoretical work has shown, the trainability of the subsequent variational circuit [13, 33].

Basis encoding maps a bit string directly onto computational basis states, requiring one qubit per bit of the (typically pre-discretized) classical input. It scales poorly for continuous, high-dimensional features and is rarely used for image-derived embeddings.

Angle (rotation) encoding assigns each classical feature x_i to a rotation angle on a dedicated qubit, e.g., $R_y(x_i)$, and therefore requires $n = N$ qubits for an N -dimensional feature vector. It has the practical advantage of a shallow, easily implementable preparation circuit and has been shown to produce more consistent results on real QPUs than more complex encodings in at least one comparative hardware study [29].

Amplitude encoding embeds a normalized vector $x \in \mathbb{R}^N$, $\|x\|_2 = 1$, into the amplitudes of an n -qubit state,

$$|\psi(x)\rangle = \sum_{i=0}^{N-1} x_i |i\rangle \in \mathcal{H}_{2^n}, \quad N = 2^n, \quad (5)$$

requiring only $n = \log_2 N$ qubits. This exponential compression is the primary motivation for its use in hybrid CNN–VQC pipelines, since it allows, e.g., a 64-dimensional PCA-reduced embedding to be encoded on just $\log_2 64 = 6$ qubits.

IQP and ZZ feature maps apply layers of classically controlled phase rotations—in particular the instantaneous quantum polynomial (IQP) circuit structure—to produce highly non-linear, problem-specific feature maps [9]. The resulting kernel $k(x, x') = |\langle \psi(x) | \psi(x') \rangle|^2$ can be exponentially hard to evaluate classically, lending potential computational interest to quantum kernel classifiers (Section 4.4). Empirically, IQP-based encodings have been observed to improve precision on some tasks while reducing recall and F_1 scores relative to angle encoding, and their performance is more sensitive to hardware noise [29].

Data re-uploading addresses the limited expressibility of a single encoding layer by repeatedly interleaving data-encoding gates with trainable rotation layers throughout the circuit, effectively allowing each qubit to “see” the input multiple times [26]. This strategy has been shown to confer universal function-approximation properties to single qubits and provides a practical route to expressible, shallow circuits compatible with NISQ depth budgets.

The compression advantage of amplitude encoding is not free: in the general case, exact amplitude-encoding state preparation for an arbitrary N -dimensional vector requires a circuit of depth $O(N)$, i.e., *exponential* in the number of qubits, unless the vector possesses exploitable structure or approximate preparation is acceptable [1]. This “input bottleneck” is a widely recognized caveat in the QML literature, and claims of exponential quantum speedup based on amplitude encoding alone should be treated with caution [1, 2]. In practice, most simulation-based studies—including the case study reviewed in Section 5—rely on software-level amplitude-embedding routines (e.g., PennyLane’s `AmplitudeEmbedding`) whose classical preprocessing cost is not charged against the “quantum” resource budget, a simplification that must be kept in mind when extrapolating simulator-based results to real hardware.

4.2 Variational Quantum Circuits

Given an encoded input state $|\psi(x)\rangle$, a VQC applies a parametrized unitary $U(\theta)$ —the *ansatz*—and measures an observable M (typically a Pauli operator) to produce a scalar prediction,

$$f(x, \theta) = \langle \psi(x) | U^\dagger(\theta) M U(\theta) | \psi(x) \rangle \in [-1, 1]. \quad (6)$$

When $M = Z_0$ (the Pauli- Z operator on the first qubit), this quantity is linearly remapped to a pseudo-probability,

$$\sigma(f) = \frac{f + 1}{2} \in [0, 1], \quad (7)$$

which can be thresholded for binary classification. A common and hardware-efficient ansatz design alternates layers of single-qubit rotations with a fixed entangling pattern: each layer $\ell = 1, \dots, L$ applies a general single-qubit rotation $\text{Rot}(\phi, \theta, \omega)$ to every qubit, followed by a ring of CNOT gates connecting qubit w to qubit $(w + 1) \bmod n$. For n qubits and L layers, this yields

$$P_{\text{VQC}} = 3nL \quad (8)$$

trainable parameters—e.g., $3 \times 6 \times 3 = 54$ rotation parameters for a 6-qubit, 3-layer circuit, plus 6 learnable input-scaling parameters and a 2-parameter measurement basis, yielding the 62 parameters of the case-study VQC—versus the 2,113 parameters of a comparable classical dense head (Equation 4). Gradients of $f(x, \theta)$ with respect to θ can be computed on real hardware via the parameter-shift rule, or, in simulation, via reverse-mode automatic differentiation [21].

4.3 Quantum Convolutional Neural Networks

Quantum convolutional neural networks (QCNNs) translate the locality and parameter sharing of classical CNNs into the quantum circuit formalism. The archetypal QCNN design, introduced by Cong et al. [5] and adapted for NISQ hardware by Wei et al. [38], alternates *quantum*

convolutional layers—parametrized two-qubit gates applied to neighboring qubit pairs—with *quantum pooling layers* that halve the active qubit register via measurements and classically controlled single-qubit operations, mimicking the spatial downsampling of classical max-pooling. A key theoretical result is that QCNs built on translational symmetry are free from the barren-plateau problem under local cost functions, providing a trainability guarantee absent from more generic parametrized circuits [27].

In the context of medical image classification, Khatoniar et al. [15] applied QCN variants to CXR pneumonia classification under extreme feature reduction, reporting competitive robustness relative to VQC baselines of comparable depth. Wei et al. [38] demonstrated digit recognition and image filtering tasks using a hardware-implementable QCN whose parameter count is *independent of input size* (determined only by circuit depth and qubit count), an attractive property for deployment on resource-constrained platforms.

It is important to distinguish QCNs from *quanvolutional* layers [11], in which a small fixed or parametrized quantum circuit processes local image patches in a sliding-window fashion, analogously to a classical convolutional kernel, before passing the resulting feature map to a standard classical CNN. Quanvolutional layers add a quantum pre-processing stage rather than replacing the classical network, and are primarily motivated by the conjecture that quantum circuits can generate feature transformations qualitatively distinct from those achievable by classical filters.

4.4 Quantum Kernel Methods

Quantum kernel methods provide an alternative to the parametrized-circuit paradigm by using a quantum circuit exclusively to evaluate a kernel function,

$$k(x, x') = |\langle \psi(x) | \psi(x') \rangle|^2, \quad (9)$$

and delegating classification to a classical support vector machine (SVM) or kernel ridge regressor trained on the resulting kernel matrix. Havlíček et al. [9] showed, using the ZZ feature map, that the corresponding quantum kernel can be classically hard to estimate exactly, providing a theoretical grounding for potential quantum advantage. Schuld and Killoran [32] established the connection between quantum models and kernel methods more broadly, showing that every VQC implicitly defines a kernel via its state overlap.

For medical imaging, the high dimensionality of image-derived feature vectors poses a practical challenge: evaluating the kernel matrix over a training set of N_{tr} samples requires $O(N_{\text{tr}}^2)$ circuit executions, each querying the state overlap of two encoded feature vectors. Haug et al. [8] proposed scalable quantum kernel estimation via randomized measurements, reducing both the circuit depth required and the sensitivity to noise, and demonstrated large-scale kernel estimation on IBM quantum hardware with noise mitigation. In a comparative hardware study, angle encoding achieved the best overall kernel performance on real QPUs among the encodings tested, albeit with pronounced run-to-run variability, while IQP feature maps performed more consistently yet underperformed relative to classical kernels; dynamical decoupling error mitigation did not uniformly improve performance and was encoding-dependent [29], underscoring that hardware-aware encoding selection is essential rather than optional.

4.5 The Barren-Plateau Problem

The central obstacle to training deep or highly entangling VQCs is the *barren-plateau* phenomenon: for a broad class of randomly initialized, sufficiently expressive ansätze evaluated against a global cost function, the variance of the cost-function gradient with respect to any single parameter θ_i decays exponentially with the number of qubits n ,

$$\text{Var}[\partial_{\theta_i} C(\theta)] = O(2^{-n}), \quad (10)$$

so that, in expectation, gradient-based optimization becomes exponentially difficult to distinguish from noise as n grows [19]. Subsequent work has elucidated the mechanisms underlying this concentration. Cerezo et al. [4] showed that the onset of barren plateaus depends on the *locality* of the cost function: global cost functions—those defined via an operator acting non-trivially on all qubits—are markedly more prone to barren plateaus than local ones. Ortiz Marrero et al. [18] demonstrated that excessive (volume-law) entanglement in the circuit is sufficient to induce barren plateaus independently of the specific ansatz structure. Kashif and Al-Kuwari [13] developed a unified framework showing that barren-plateau onset depends jointly on data encoding, ansatz expressibility, and entanglement, providing design constraints for HQC networks.

These theoretical results have motivated several practical mitigation strategies:

- **Local cost functions and shallow entangling ansätze.** Restricting measurements to local Pauli observables and using nearest-neighbor entangling layers (e.g., the CNOT ring in Equation 8) substantially suppresses gradient variance [4].
- **Controlled initialization.** Initializing parameters near the identity, or near simple product states (“identity blocks”), rather than uniformly at random, ensures that the optimization starts in a region of non-vanishing gradient [19, 25]. Partitioning the circuit into cost and non-cost registers and using low-entanglement initializations provides a practical recipe [25].
- **Expressibility-aware circuit design.** Reducing circuit expressibility—via state-efficient ansätze (SEA) [17] or neural-network-based parameter generation that smooths the loss landscape [40]—trades representational power for trainability.
- **Noise-assisted mitigation.** Entanglement regularization, the addition of controlled Langevin noise, and rotations into the cost function’s eigenbasis can improve convergence on difficult optimization landscapes [25].
- **Noise mitigation as expressibility reduction.** Ding et al. [7] showed that adapting circuit expressibility to match the noise level of the hardware can act as an error-aware mitigation strategy, improving classification accuracy in noisy simulations for QCNNs.

These design constraints guided the layer-count selection for the case-study ansatz in Section 5: three data re-uploading layers on six qubits keep the circuit in a regime of moderate expressibility and shallow entanglement, where gradient-variance concentration is not expected to dominate training on a noise-free simulator.

4.6 On the Prospect of Quantum Advantage

A recurring question in the QML literature is whether VQC-based classifiers can offer a genuine *computational* advantage—in runtime, sample complexity, or expressivity—over classical models, as opposed to merely a reduction in the number of trainable parameters. At the circuit sizes considered in this review (single-digit to low tens of qubits), no such advantage has been established for medical image classification: circuits of this size are efficiently simulable classically, and the reduction in decision-layer parameter count does not, by itself, imply reduced overall computational cost, since the classical feature-extraction and dimensionality-reduction stages typically dominate the total pipeline cost. Claims of quantum advantage in QML more broadly remain an active and contested research area, further complicated by the input/output bottleneck discussed above [1, 2]. The practical value of hybrid VQC decision layers at present should therefore be understood primarily in terms of *parameter efficiency* and *memory footprint*—properties that may be relevant for edge or resource-constrained deployment once fault-tolerant or larger-scale NISQ hardware becomes available—rather than as evidence of an accuracy or speed advantage over well-tuned classical models.

5 Hybrid Quantum–Classical Architectures

5.1 Design Rationale

Given the qubit and depth constraints of NISQ hardware (Section 4) and the high dimensionality of raw medical images, a natural design principle is to separate *feature learning*, delegated to a classical CNN operating on the full-resolution image, from *decision making*, delegated to a compact quantum circuit operating on a low-dimensional, information-dense representation. This principle is shared by all hybrid architectures reviewed below, differing primarily in (i) the choice of classical backbone, (ii) the dimensionality reduction method, (iii) the encoding and ansatz design, and (iv) whether the quantum component serves as a classifier head, a convolutional feature transformer, or a kernel function.

5.2 Related Work on Hybrid Quantum–Classical CXR Classifiers

Table 1 summarizes published hybrid QML studies targeting chest X-ray analysis. Three themes emerge from this survey.

Table 1: Representative hybrid quantum–classical studies on CXR or pulmonary imaging tasks. All results are from noise-free simulators unless marked with † (IBM Q backend) or ‡ (this work additionally reports execution on physical quantum hardware). “HQC-VQC” denotes a classical feature extractor followed by a VQC classifier head; “HQC-Conv” denotes a quantum convolutional layer integrated into the CNN pipeline.

Reference	Architecture	Dataset	Task	Key result
Kulkarni et al. [14]	HQC-VQC	CXR (Guangzhou)	Pneumonia	Stat. significant improvement vs. classical CNN baseline
Rao et al. [30]	HQC-VQC (†)	COVID-19 Radiography Dataset (15K)	4-class	98.1% test acc.; IBM Q QASM deployment
Decoodt et al. [6]	HQC-VQC	CheXpert (2.4K PA)	Cardiomegaly	$AUC \leq 0.93$; improved Grad-CAM++ localization
Sharma [36]	HQC-Conv	Custom CXR	Pneumonia	ZFNet + quantum layer hybrid
Khatoniar et al. [15]	QCNN	CXR (Guangzhou)	Pneumonia	Robustness analysis under extreme feature reduction
This work	HQC-VQC (‡)	CXR (Guangzhou, 5.9K)	Pneumonia	Bal. accuracy 0.747 (VQC) vs. 0.729 (MLP), $p = 0.71$; $34\times$ param. reduction; on-device noise study

Consistent accuracy–efficiency trade-off. All surveyed studies confirm the qualitative finding that a hybrid quantum component can achieve meaningful classification performance at a substantially reduced decision-layer parameter budget. However, results that appear favorable relative to their specific classical baselines often reflect comparisons against purposely weak baselines rather than state-of-the-art classical models [1].

Variability in experimental rigor. Dataset sizes range from a few hundred to $\sim 15,000$ images; train/test split rationale is frequently underspecified; and class prevalence in the test

set is rarely reported explicitly. This limits direct cross-study comparison and motivates the shift-aware evaluation framework presented in Section 7.

Hardware deployment and the missing simulator–hardware comparison. Among the surveyed studies, only Rao et al. [30] report deployment on an IBM Q backend, and that deployment targets a QASM simulator rather than physical qubits; none quantifies the gap between noise-free simulation and hardware execution for the same trained circuit. Section 6 addresses this gap directly for the pipeline of Section 5, reporting both the magnitude of the hardware-induced degradation and a gate-level attribution of its cause.

5.3 Pipeline Architecture

The pipeline consists of four stages, summarized in Table 2: (1) a frozen ConvNeXt-Tiny backbone, used headless, maps each 224×224 input image to a 768-dimensional embedding; (2) a domain-adversarial neural network (DANN) with a gradient reversal layer (GRL) is trained on top of the embedding to suppress the dataset shift between the training and test distributions (Section 5.2); (3) a nonlinear autoencoder, fitted on the training embeddings, reduces this embedding to 64 dimensions, chosen to match the Hilbert-space dimension of a 6-qubit register ($2^6 = 64$); and (4) a 6-qubit, 3-layer VQC produces a scalar pneumonia score via Equations 6–7. The circuit *amplitude-encodes* the 64-dimensional latent vector once (Equation 5), then applies $L = 3$ data re-uploading layers, each consisting of scaled single-qubit rotations $R_y(\text{scale}_q \cdot x_q \cdot \pi)$ that re-upload only the first six latent components $x_0, \dots, x_5$ (the re-uploading budget of a 6-qubit register), the general rotation $\text{Rot}(\phi, \theta, \omega)$ of Section 4.2, and a ring of CNOT entanglers. The remaining 58 latent dimensions reach the classifier exclusively through the single amplitude embedding; this explicit trade-off is revisited in Section 8.

Table 2: Hybrid pipeline architecture and parameter counts.

Stage	Input	Operation	Output	Parameters
Feature extraction	224×224 image	ConvNeXt-Tiny (frozen)	768-D vector	0 (fixed)
Domain adaptation	768-D	DANN + GRL	768-D vector	0 (fixed)
Dimensionality reduction	768-D	Autoencoder (nonlinear)	64-D vector	0 (fixed)
Quantum classification	64-D	VQC $U(\theta)$	score $\in [0, 1]$	62

Among linear PCA, linear discriminant analysis (LDA), and univariate feature selection (SelectKBest), a nonlinear autoencoder yielded the most stable training dynamics and the best validation performance at 64 latent dimensions; LDA suffered from numerical instability in the high-dimension/limited-sample regime, and SelectKBest achieved marginally weaker validation metrics. The autoencoder therefore replaces the linear projection used in earlier variants of this pipeline.

5.4 Dataset and Preprocessing

The pipeline was evaluated on the publicly available pediatric CXR pneumonia dataset originating from the Guangzhou Women and Children’s Medical Center (5,856 anteroposterior CXR images, ages approximately 1–5 years). The original release provides a validation split of only 16 images, which is insufficient for reliable convergence monitoring; the training and validation partitions were therefore merged and re-split 80:20 with class stratification using a *patient-grouped* shuffle, so that no patient contributes images to both sides of the split, while the original test

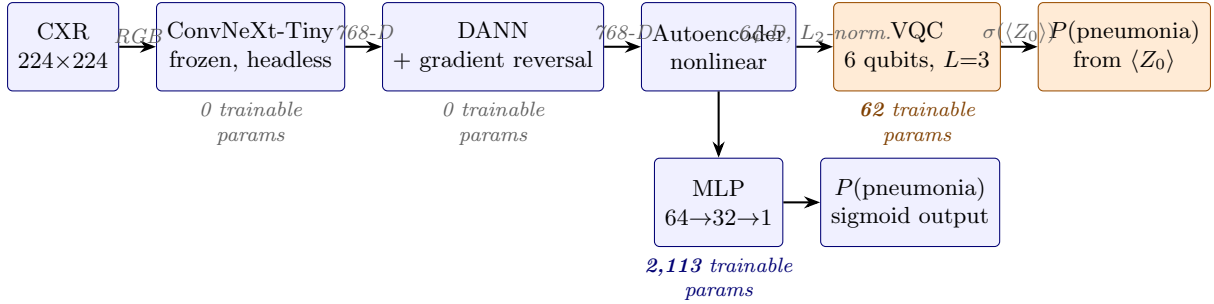


Figure 1: Schematic of the hybrid pipeline. A frozen ConvNeXt-Tiny embeds each 224×224 radiograph into a 768-dimensional vector; a domain-adversarial stage with a gradient reversal layer suppresses the train/test distribution shift, and a nonlinear autoencoder compresses the embedding to 64 dimensions. The compressed vector is then classified either by the 6-qubit, 62-parameter data re-uploading VQC (upper branch, this work) or by the 2,113-parameter MLP baseline (lower branch), both operating on identical features. The 64-dimensional latent is chosen to match the Hilbert-space dimension of a 6-qubit register ($2^6 = 64$).

partition was left untouched to preserve the integrity of the final evaluation. The resulting split is summarized in Table 3. Notably, the test set exhibits a materially different class prevalence (62.5% pneumonia) than the training and validation sets (74.4% and 73.3% respectively), constituting an explicit, quantified dataset shift of roughly 11 percentage points that is exploited below as a robustness stress test rather than concealed.

Table 3: Dataset composition and class-prevalence shift. The train/validation re-split is patient-grouped; the test partition is the original, patient-disjoint split, unmodified.

Split	Images	Pneumonia (%)	Normal (%)
Training	4,243	74.4	25.6
Validation	989	73.3	26.7
Test	624	62.5	37.5

Standard image preprocessing was applied prior to the ConvNeXt-Tiny forward pass: resizing to 224×224 pixels, conversion to a normalized float tensor, and channel-wise standardization using ImageNet mean $[0.485, 0.456, 0.406]$ and standard deviation $[0.229, 0.224, 0.225]$, matching the statistics of the pretraining distribution. Because the CXR images are grayscale while the backbone expects three input channels, the single channel is duplicated across the RGB channels. Medical-safe augmentations were limited to mild rotations ($\pm 7^\circ$), small affine translations, and modest color jitter; aggressive augmentations such as horizontal flipping and RandAugment were deliberately excluded as they are harmful for anatomical X-ray content.

5.5 Training Configuration and Loss Functions

The VQC and a classical baseline classifier (a single-hidden-layer MLP with 32 hidden units, operating on the same 64-dimensional autoencoder features) were trained independently, each with the Adam optimizer, a batch size of 16, and early stopping (patience of 3 epochs) on validation loss. Because VQC outputs (Equation 6) lie in $[-1, 1]$ rather than $[0, 1]$, binary labels $y_i \in \{0, 1\}$ were remapped to targets $t_i \in \{-1, +1\}$, and the quantum classifier was trained with a class-weighted mean squared error,

$$\mathcal{L}_{\text{QMSE}} = \frac{1}{N} \sum_{i=1}^N w(t_i) (z_i - t_i)^2, \quad (11)$$

where z_i is the raw circuit output and $w(\cdot)$ up-weights the minority (Normal) class to counteract the training-set prevalence imbalance. The classical baseline was trained with a class-weighted binary cross-entropy loss on its sigmoid output. The quantum classifier’s learning rate followed a linear warm-up over the first $E_{\text{warmup}} = 3$ epochs followed by cosine annealing,

$$\eta(e) = \begin{cases} \eta_0 \frac{e}{E_{\text{warmup}}}, & e \leq E_{\text{warmup}}, \\ \eta_{\min} + \frac{1}{2}(\eta_0 - \eta_{\min}) \left[1 + \cos\left(\pi \frac{e - E_{\text{warmup}}}{E - E_{\text{warmup}}}\right) \right], & e > E_{\text{warmup}}, \end{cases} \quad (12)$$

with $\eta_0 = 10^{-3}$ and $\eta_{\min} = 10^{-5}$, while the classical baseline used unmodified cosine annealing. Gradient computation for the quantum circuit used the adjoint differentiation method [21], which recovers the gradient of a scalar observable from cached forward executions at a cost scaling linearly in the number of trainable parameters, and which is roughly two orders of magnitude faster than the parameter-shift rule on the statevector simulator used here; the parameter-shift rule remains the appropriate choice when targeting physical hardware, where execution caching is unavailable. All quantum-circuit training and classical simulation reported in this section was performed on a noise-free statevector simulator, so the results characterize the idealized, decoherence-free behavior of the ansatz; Section 6 reports the corresponding physical-hardware behavior separately. In practice both models converged well before the 50-epoch budget under early stopping.

5.6 Results

Table 4 summarizes test-set performance for the classical baseline and the hybrid VQC classifier at the validation-selected decision thresholds (Section 7.2): $\tau^* = 0.525$ for the MLP and $\tau^* = 0.800$ for the VQC. Both thresholds were selected to maximize balanced accuracy on the validation set only, and applied once to the test set. Both models see identical features from an identical, patient-disjoint split, so the comparison isolates the effect of the decision layer.

Table 4: Test-set diagnostic performance: classical MLP baseline vs. hybrid VQC classifier (624 test images, 62.5% pneumonia prevalence). Thresholds τ^* were selected on the validation set via balanced-accuracy maximization and applied once to the test set.

Metric	Classical (MLP)	Hybrid (VQC)
Accuracy	74.20%	74.84%
Balanced accuracy	0.7295	0.7466
Precision	80.21%	82.82%
Sensitivity (Recall)	77.95%	75.38%
Specificity	67.95%	73.93%
F_1 -score	0.7906	0.7893
AUC-ROC (test)	0.8476	0.8055
Decision threshold τ^*	0.525	0.800
Decision-layer parameters	2,113	62
Parameter reduction	$\approx 34\times$	

The two models are statistically indistinguishable but differ in profile. The VQC attains the higher test accuracy (74.84% versus 74.20%), the higher balanced accuracy (0.7466 versus 0.7295), the higher precision (82.82% versus 80.21%) and substantially the higher specificity (73.93% versus 67.95%), at the cost of lower sensitivity (75.38% versus 77.95%), a marginally lower F_1 -score (0.7893 versus 0.7906), and a lower threshold-independent ranking quality (test AUC-ROC 0.8055 versus 0.8476). An exact McNemar test on the 624 paired test predictions

yields $p = 0.71$, so the accuracy difference is not significant at $\alpha = 0.05$; non-parametric bootstrap resampling ($B = 1000$) likewise returns overlapping 95% confidence intervals for AUC-ROC, $[0.818, 0.877]$ for the MLP against $[0.766, 0.844]$ for the VQC. We therefore characterize the hybrid model as *comparable* to the classical baseline, and explicitly do not claim that it is superior: the MLP ranks the test set better, and the VQC’s advantage is confined to the thresholded operating point selected by the balanced-accuracy criterion. The notable positive result is that this operating point is reached with approximately $34\times$ fewer trainable decision-layer parameters.

The two selected thresholds differ sharply ($\tau^* = 0.525$ against 0.800), which is instructive rather than incidental. Because Equation 7 maps $\langle Z_0 \rangle$ linearly onto $[0, 1]$, the VQC’s scores are not calibrated against the sigmoid MLP’s scores, so the two models cannot be thresholded at a common τ and must instead be calibrated separately on the validation set. The VQC’s selected $\tau^* = 0.800$ is a markedly conservative operating point, consistent with its higher specificity and lower sensitivity: the circuit concentrates its outputs in the upper region of $[0, 1]$ and requires a correspondingly high cut to balance the two error types. This calibration mismatch is a practical cost of the hybrid design that is easily overlooked when a quantum head is treated as a drop-in replacement for a dense layer, and it follows directly from fitting the $\{-1, +1\}$ -targeted objective of Equation 11 in a space that is only afterwards affinely remapped to a probability.

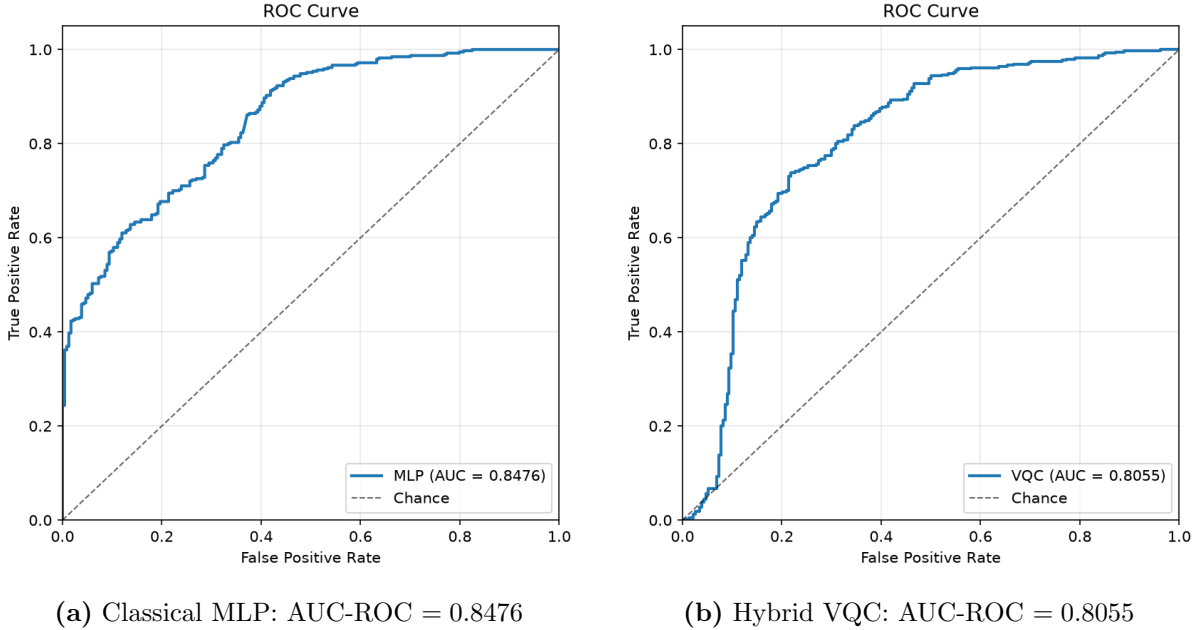


Figure 2: Receiver operating characteristic curves for the two decision layers on the 624-image test set, evaluated on identical features. The MLP attains the higher area under the curve, consistent with the bootstrap intervals of Section 5 overlapping across $\alpha = 0.05$.

Compared to the literature surveyed in Table 1, the balanced accuracies reported here are considerably lower than the near-perfect figures reported by Rao et al. [30] or Kulkarni et al. [14]. We attribute this to: (i) the explicit train/test class-prevalence shift applied here; (ii) a single-run evaluation without cross-validation; (iii) the absence of fine-tuning of the ConvNeXt-Tiny backbone; and (iv) the deliberately small quantum decision layer, whose 62 parameters limit the capacity available to model the 64-dimensional latent space. These factors collectively reduce the upper bound on achievable performance and make the comparison more conservative and, arguably, more realistic: the comparison of interest in this paper is against a competently trained classical baseline on identical features, not against the headline figures of studies using different splits, different backbones and, in several reported cases, weaker reference classifiers.

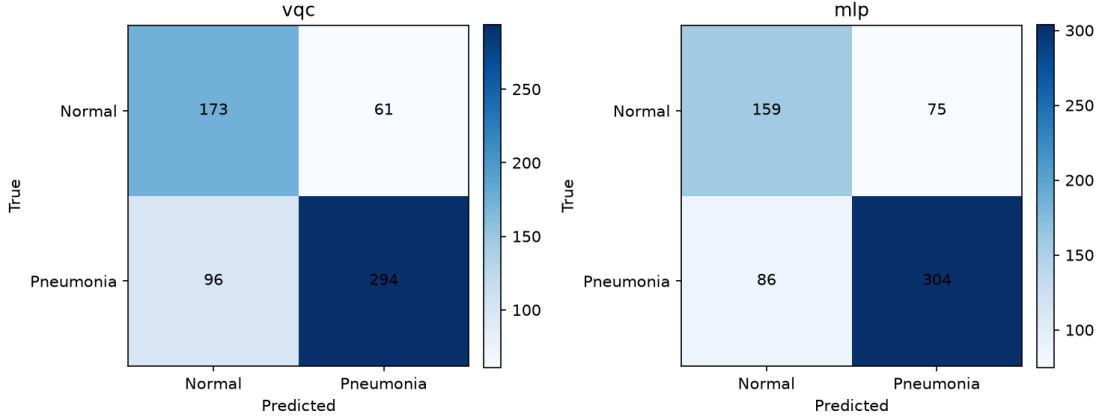


Figure 3: Test-set confusion matrices at the validation-selected thresholds, $\tau^* = 0.525$ for the MLP (left) and $\tau^* = 0.800$ for the VQC (right). Both models err more often by over-flagging normal studies as pneumonia than by missing pneumonia cases, a direct consequence of the prevalence shift documented in Table 3: both were trained under a roughly 74% pneumonia prior and are evaluated at 62.5%.

6 Experiments on Real Quantum Hardware

This section reports the execution of the trained VQC of Section 5 on physical quantum hardware, and quantifies the resulting simulator-to-hardware gap. We report the outcome as a negative result. On the hardware tested, noise dominates the circuit’s output to the point where the trained classifier’s discriminative behaviour is no longer recoverable, and we show by gate accounting that this failure is attributable primarily to the state-preparation stage rather than to the learned ansatz.

6.1 Motivation and Hardware Selection

Experiments were performed on `ibm_miami`, a 127-qubit superconducting processor accessed through the Qiskit Runtime `SamplerV2` primitive. Jobs were deliberately submitted outside an execution *session*: sessions bill wall-clock time irrespective of device occupancy, which is an inappropriate accounting model for an experiment whose queue latency dominates its compute time. Circuits were transpiled to the device’s native basis set (`{ECR, RZ, SX, X}`) at optimization level 3, with the 6-qubit ansatz mapped onto a short chain of connected, low-error qubits to avoid SWAP insertion. All circuits were sampled with 1,024 shots. We use this as an integrity check throughout: every expectation value recovered from hardware in our logs is an exact multiple of $1/1024$, which distinguishes genuine sampled data from simulator output.

The transpiled two-qubit gate accounting of Table 5 is the central diagnostic for interpreting the results. The asymmetry is severe. The amplitude encoding of Equation 5 requires $2^6 - 2 = 62$ CNOT gates under the standard Möttönen decomposition [22], whereas the entire trainable three-layer ansatz contributes only $3 \times 6 = 18$. State preparation therefore accounts for roughly 78% of the circuit’s two-qubit gate count—about $3.4\times$ the depth of the learned circuit itself. The trainable parameters constitute under a quarter of the executed circuit, which means that parameter-efficient ansatz design buys comparatively little on hardware: what dominates the error budget is a data-loading routine whose depth grows exponentially with the number of encoded amplitudes and which no amount of ansatz tuning can offset.

Table 5: Transpiled two-qubit (CNOT) gate budget for the 6-qubit circuit.

Circuit stage	CNOTs	Share
Amplitude encoding (state preparation)	62	77.5%
Trainable ansatz ($L = 3$ data re-uploading)	18	22.5%
Measurement basis	0	0%
Total	80	100%

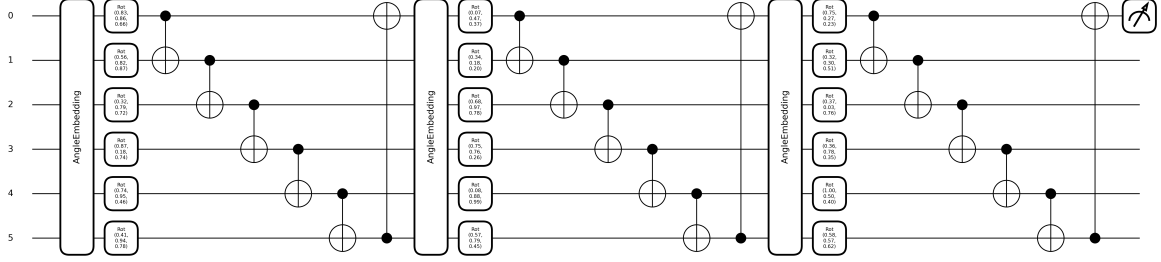


Figure 4: The trained ansatz transpiled to the target device’s native gate set at optimization level 3. The circuit is visually dominated by the amplitude-encoding prefix, consistent with the gate budget of Table 5; the trainable layers constitute a small fraction of the total depth.

6.2 Error Mitigation Strategies

We applied zero-noise extrapolation (ZNE) using Richardson extrapolation [20] with noise scale factors $\{1, 2, 3\}$, implemented by gate folding. For each sample the three scaled circuits were submitted as a single batched job, so mitigation triples the shot cost without tripling queue latency. Dynamical decoupling and Pauli twirling were additionally applied to suppress uncorrelated errors.

The mitigation result is instructive precisely because it is only partial. Table 6 shows that ZNE *restores* the distributional span that noise destroyed—from 0.151 to 0.554—but overshoots the ideal span of 0.404 by 37%. Two diagnostics make the instability explicit. First, the sample-level Pearson correlation between mitigated and ideal predictions falls to $r = 0.07$, against $r = 0.70$ for the raw hardware predictions: the mitigation preserves the aggregate distribution while destroying the agreement on individual samples. Second, the overshoot itself indicates that the linear model of the noise scaling parameter assumed by Richardson extrapolation does not hold over the fitted range. ZNE at this circuit scale therefore mitigates the *variance* of the output distribution while introducing additional *bias* in the individual predictions. This is consistent with the comparative hardware literature, in which mitigation effectiveness is strongly encoding-dependent and no universally applicable protocol has emerged [29, 8].

We stress one methodological caveat about the reference row of Table 6. The noisy-simulation reference was produced with a FakeKingston backend, i.e. a snapshot of the calibration data of a *different* processor (`ibm_kingston`) from the one on which the samples were actually executed (`ibm_miami`). That row is therefore an indicative noise reference rather than a controlled same-device simulation, and the gap between it and the raw hardware row conflates device-specific noise structure with the effect of mitigation. A controlled comparison would require re-running the identical circuit on a single device under both conditions; we report the available data as measured and flag the limitation rather than presenting the two rows as a like-for-like decomposition.

Table 6: Noise ladder for a class-balanced 10-sample subset: distributional span ($\max - \min$) of the mapped pneumonia probability $p = (1 + \langle Z_0 \rangle)/2$ under noise-free simulation, noisy simulation, raw hardware, and ZNE-mitigated hardware. Note that the noisy reference comes from a FakeKingston snapshot of a different processor (`ibm_kingston`) than the hardware used for the raw and mitigated rows (`ibm_miami`); see the caveat above.

Condition	Span of p	Relative to ideal
Ideal statevector simulation	0.404	—
Noisy simulation (FakeKingston)	0.285	−29%
Raw hardware, <code>ibm_miami</code> (1,024 shots)	0.151	−62%
ZNE-mitigated hardware	0.554	+37%

6.3 Hardware Results and Simulator–Hardware Gap

Two hardware campaigns were executed. The first comprised 79 batched jobs run on 15 September 2026, and was independently audited against the provider’s workload manifest: all 79 job identifiers appear in the IBM Quantum workload record for the account, with 79 jobs completed and one cancelled. This campaign exposed a defect in our own evaluation harness rather than in the circuit—the hardware subset had been drawn by sorting on the label column, which produced an all-negative slice and consequently rendered any accuracy or AUC-ROC estimate undefined. We report it as an audit record only, and draw no performance conclusion from it. The defect was corrected by drawing the hardware subset with stratified, class-balanced sampling, and the second campaign re-ran the protocol on 50 class-balanced test images (25 pneumonia, 25 normal).

On the corrected subset the trained circuit fails outright. The ideal simulator assigns these 50 samples a probability span of 0.439; the hardware assigns them a span of 0.182, a collapse of 59%. The raw hardware predictions occupy the interval $[0.445, 0.627]$ with mean 0.528—a narrow band around chance that largely discards the simulator’s ranking. No threshold applied to this distribution yields a classifier whose accuracy we would be willing to report.

We read this outcome as a confirmation of the state-preparation analysis in Table 5 rather than as evidence about the quality of the learned parameters. A circuit in which 77.5% of the two-qubit gates belong to an $O(N)$ state-preparation routine—never optimized, trained, or error-compensated, and exponential in the register size—will accumulate an error budget dominated by data loading; a 62-parameter ansatz cannot be expected to survive that. This also explains the asymmetry flagged in Section 4.1: simulation-based studies that rely on a software-level amplitude-embedding primitive effectively assume state preparation is free, and thereby measure only the component of the pipeline that hardware penalizes least. Our own simulator results in Table 4 inherit exactly this idealization, which is why we regard them as an upper bound on achievable device performance.

Two practical consequences follow. First, for amplitude-encoded VQCs on present hardware the design variable that governs achievable accuracy is encoding depth, not trainable parameter count; approximate and distributed state preparation [1] is the lever with real leverage, whereas reducing the ansatz from three layers to one spends expressibility for no hardware benefit. Second, error mitigation evaluated at the level of a distribution summary—as in Table 6—can appear to succeed while per-sample fidelity collapses. Any credible mitigation claim must therefore be reported at the level of individual predictions, not of aggregate spread.

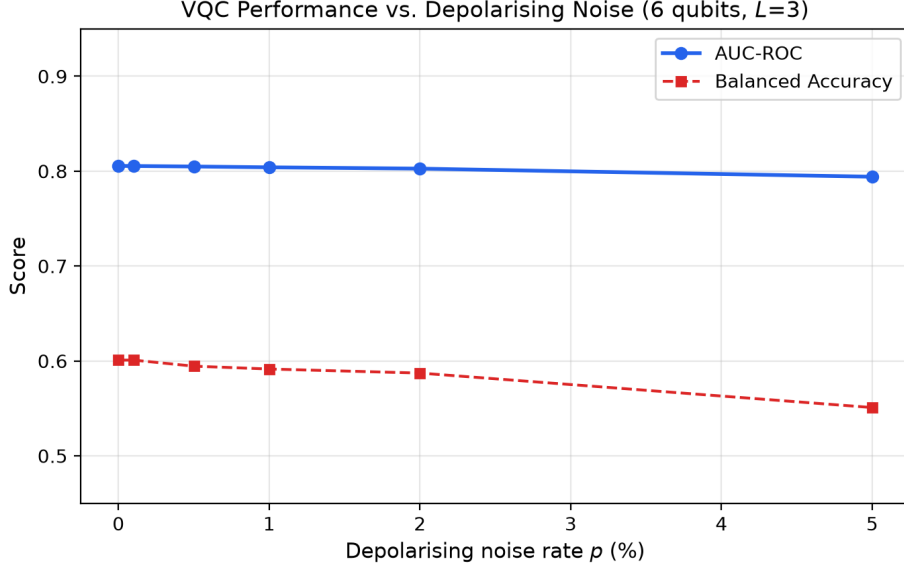


Figure 5: Simulated depolarizing-noise ablation, distinct from the FakeKingston reference of Table 6. Test AUC-ROC of the hybrid VQC under increasing depolarizing rate on a mixed-state simulator, from the noise-free operating point (AUC 0.8055) to $p_{\text{noise}} = 0.05$. The shallow degradation observed in simulation contrasts sharply with the collapse measured on hardware (Section 6.3), indicating that a single depolarizing parameter tuned to match one region of the noise landscape does not reproduce the correlated, gate-dependent errors of a real device.

7 Experimental Considerations and Evaluation Metrics

7.1 Metrics

Given the class imbalance typical of public pneumonia CXR datasets, accuracy alone is an insufficient evaluation criterion. We recommend reporting, at minimum: sensitivity (recall), $\text{Sens} = TP/(TP + FN)$; specificity, $\text{Spec} = TN/(TN + FP)$; balanced accuracy, $\text{BalAcc} = \frac{1}{2}(\text{Sens} + \text{Spec})$; the F_1 -score; and the area under the receiver operating characteristic curve (AUC-ROC), which is threshold-independent and therefore particularly informative when comparing classifiers (such as a VQC and an MLP) whose raw outputs have different natural ranges and calibration properties.

7.2 Threshold Selection

Because the VQC output (Equation 7) and the classical sigmoid output are continuous scores, a decision threshold τ must be chosen to obtain a binary prediction $\hat{y} = \mathbb{I}[p_{\text{pneumonia}} > \tau]$. A principled approach is to perform a discrete threshold scan on the validation set, selecting $\tau^* = \arg \max_{\tau} \text{BalAcc}(\tau)$, and to report test-set metrics at this fixed, validation-selected threshold rather than at an arbitrarily chosen value. In the case study reviewed here, this procedure selected $\tau^* = 0.525$ for the MLP and $\tau^* = 0.800$ for the VQC over a scan range of $\tau \in [0.30, 0.80]$ with step 0.025. The two selected thresholds are strikingly different, and the reason is instructive rather than incidental: the remapping of Equation 7 places the VQC’s scores in the upper region of $[0, 1]$, so the VQC is calibrated to a conservative, high-specificity operating point, whereas the MLP selects a threshold near the conventional default and retains the higher sensitivity reported in Table 4. Threshold sensitivity analysis remains a useful diagnostic in both cases: near $\tau = 0.5$ the MLP’s operating point shifts toward higher specificity at the cost of sensitivity, whereas the balanced-accuracy criterion selects the VQC’s high-threshold regime. Reporting AUC-

ROC alongside thresholded metrics is therefore important, since a single threshold-dependent operating point can misrepresent the relative quality of two models whose output scales are not comparable.

7.3 Dataset Shift as an Explicit Evaluation Axis

We highlight dataset shift as a first-class evaluation concern rather than an incidental property of the data split. Reporting class prevalence separately for training, validation, and test partitions (as in Table 3) allows a reader to distinguish genuine generalization performance from performance inflated by matching train/test class ratios. This is particularly relevant for QML studies, where sample-efficiency constraints (imposed by simulator run-time or hardware queue availability) often preclude the large-scale cross-validation protocols that are standard in classical deep learning benchmarking; a single, explicitly characterized shift-aware split is a partial, low-cost mitigation. The generalizability limitations of single-institution, narrow-demographic datasets highlighted by Zech et al. [39] apply with equal force to hybrid quantum–classical architectures.

7.4 Interpretability

For classical CNN components, gradient-based saliency methods such as Grad-CAM can verify that the network attends to clinically relevant lung regions rather than to background artifacts (e.g., imaging equipment, laterality markers), providing a qualitative sanity check complementary to aggregate performance metrics. Decoodt et al. [6] found that augmenting a DenseNet-121 backbone with a variational quantum layer improved Grad-CAM++ localization quality for cardiomegaly detection, suggesting a potential ancillary benefit of quantum components beyond raw classification accuracy. Analogous interpretability tools for the quantum component are considerably less mature; feasible directions include measuring the sensitivity of the circuit output to perturbations of individual input amplitudes, or analyzing feature importance in the classical dimensionality-reduction stage (e.g., via SelectKBest or PCA loading vectors) as a proxy for what information reaches the quantum classifier.

8 Challenges, Limitations, and Open Problems

We summarize the principal limitations of current hybrid VQC approaches to CXR pneumonia detection, several of which are directly evidenced by the case study in Section 5.

1. **Barren plateaus.** As circuit width or entangling depth increases, gradient variance decays exponentially (Equation 10), limiting the ansätze that can be reliably trained with current optimizers [19, 4, 18]. The case-study ansatz was therefore deliberately restricted to three shallow layers on six qubits, a regime in which gradient-variance concentration is not expected to dominate training (Section 4.5). While several mitigation strategies exist, none fully eliminates the fundamental exponential scaling [13].
2. **State-preparation and encoding overhead.** Amplitude encoding achieves exponential qubit compression but, in the general case, incurs exponential state-preparation circuit depth [1]; simulator-based studies typically do not account for this cost. In the present pipeline it is not a caveat but the dominant cost: encoding a real-valued 64-dimensional vector into six qubits requires $2^6 - 2 = 62$ CNOT gates under the standard Möttönen decomposition [22], against only 18 CNOTs (3 layers $\times$ 6 qubits) in the entire trainable ansatz, so state preparation alone accounts for roughly 78% of the circuit’s two-qubit gates. We confirm on hardware that this budget dominates in practice: the measured collapse in output span reported in Section 6.3 is attributable primarily to this stage rather than to the learned

parameters. Our simulator results should consequently be read as an upper bound on device performance.

3. **Dataset shift, with its effect left unquantified.** Differences in class prevalence (and, more broadly, in acquisition device, patient demographics, and imaging protocol) between training and deployment distributions are a well-documented source of generalization failure for CXR classifiers [39]. The present split deliberately retains an approximately 11-percentage-point prevalence shift (Table 3) as a stress test, and both confusion matrices in Figure 3 show over-flagging of normal studies, which is the error direction one would expect under a reduced positive prior. We note, however, that we do not isolate the shift’s effect: without a prevalence-matched control split we cannot attribute any part of the reported performance to the shift rather than to the other differences enumerated in Section 5, and the domain-adversarial stage included in the pipeline was not ablated against a no-adaptation control, so its contribution is likewise qualitative.
4. **Noise-dominant hardware behaviour at small scale.** All training and classical simulation results reported here were obtained on noise-free statevector simulators. On the physical processor tested, hardware noise reduced the span of the predicted pneumonia probability by 59% relative to simulation and collapsed the output into a narrow band around chance (Section 6.3), and zero-noise extrapolation restored the distributional span without restoring per-sample agreement with the simulator (Table 6). We report this as a negative result for this circuit, this encoding and this device—not as a general claim that hybrid quantum–classical pipelines are unviable. A single device, a single encoding and a single calibration epoch cannot support such a generalization, and our design does not attempt to control for any of them; a sweep across device families and encodings is needed before this result can be either generalized or overturned.
5. **Information loss under dimensionality reduction.** Compressing a 768-dimensional CNN embedding to 64 (or fewer) dimensions via the nonlinear autoencoder necessarily discards variance, some fraction of which may be diagnostically relevant. Alternative encodings that avoid aggressive compression (e.g., data re-uploading [26] with angle encoding, or distributed amplitude encoding over multiple shorter circuits) remain open design questions [35].
6. **Absence of demonstrated quantum advantage.** At the circuit scales considered (single-digit qubits), no computational or statistical advantage over classical models has been established; the case for VQC decision layers currently rests on parameter efficiency rather than accuracy or speed.
7. **Limited statistical robustness.** The case study reviewed here uses a single train/validation/test split without cross-validation or repeated randomized resampling, owing to the computational cost of circuit simulation. We mitigate this with an exact McNemar test ($p = 0.71$) and non-parametric bootstrap confidence intervals ($B = 1000$), the latter yielding overlapping 95% intervals for test AUC-ROC (Section 5). A paired design with repeated splits would nonetheless be required to separate the observed 0.64-percentage-point accuracy difference from run-to-run variance, and we make no claim that the difference is meaningful in either direction. Relatedly, the hyperparameter search that fixed the 64-dimensional latent size and the three-layer depth was conducted on the validation set of this same split, so the reported validation-selected quantities are mildly optimistic as estimates of generalization.
8. **Patient grouping is reconstructed, not native.** The train/validation re-split is performed at the *patient* level using a grouped shuffle, so that no patient contributes images to both sides—the leakage mode that would otherwise inflate validation metrics. Two residual

caveats remain. First, the public Kaggle distribution of the dataset does not expose patient identifiers in the image filenames, so patient identity was inferred from the structure of the source release rather than read from an authoritative metadata table; any image whose identifier does not follow the expected convention may therefore be mis-grouped. Second, the held-out test set follows the original authors’ patient-disjoint partition and was never touched by threshold optimization, so it remains the more conservative of the two estimates. A full patient-grouped re-evaluation in which the test partition is also regenerated under our own grouping would be the stronger check, and is listed as future work in Section 9.

9. **Single-institution, narrow-demographic data.** The underlying dataset originates from a single pediatric population at one medical center, limiting claims of generalizability to adult populations, other acquisition protocols, or other institutions.
10. **Unfair classical baselines.** Several studies surveyed in Table 1 compare a quantum model favorably against a deliberately weak classical reference, inflating the apparent advantage of the hybrid approach. Rigorous evaluation requires comparison against optimized classical baselines of equal or greater parameter budget on the quantum side [1].

9 Future Directions

Several directions appear promising for advancing hybrid quantum–classical pneumonia detection beyond the proof-of-concept stage.

- **Encoding depth as the primary hardware design variable.** Our hardware experiment (Section 6.3) indicates that, for amplitude-encoded circuits, achievable on-device accuracy is governed by the depth of the state-preparation routine rather than by the trainable parameter count. The most promising concrete directions are therefore approximate and distributed state preparation, which trade exact encoding for logarithmic depth [1]; angle encoding, which is shallow but qubit-expensive [29]; and re-uploading schemes that avoid a single deep preparation stage altogether. Each should be compared at matched *total circuit depth* rather than at matched parameter count, which is the comparison our own results suggest matters most.
- **Broader hardware evaluation.** A single device and a single encoding cannot support general conclusions about the simulator-to-hardware gap. Systematic sweeps across device families (superconducting and trapped-ion), qubit counts and calibration epochs—with the per-stage error attribution attempted in Table 5—are needed before the negative result reported here can be generalized or overturned. Larger hardware subsets are equally necessary: the corrected campaign in Section 6.3 used 50 class-balanced images, which is sufficient to demonstrate collapse but not to estimate accuracy.
- **Quantum convolutional and quanvolutional architectures.** QCNNs [5, 38] that exploit spatial locality in the quantum circuit, and quanvolutional pre-processing layers [11] that apply quantum filters to image patches prior to classical CNN processing, represent theoretically motivated alternatives to the “quantum classifier head” paradigm. Their trainability guarantees [27] and parameter independence from input size make them particularly attractive for deployment under NISQ constraints.
- **Quantum kernel methods.** Replacing the VQC classifier with a quantum kernel SVM [9, 32] has the advantage of convex optimization—eliminating barren plateaus by construction—and a well-defined notion of kernel quality via the quantum Fisher information [8]. Scalable kernel estimation via randomized measurements [8] makes this approach computationally feasible even for training sets of several thousand images.

- **Data re-uploading classifiers.** Applying the data re-uploading paradigm [26] within the VQC decision layer would increase circuit expressibility without requiring additional qubits, potentially alleviating the information-loss constraints imposed by aggressive dimensionality reduction to the 64-dimensional amplitude-encoded register.
- **Systematic hyperparameter optimization.** Bayesian optimization frameworks (e.g., Optuna) applied jointly across classical and quantum hyperparameters (learning rate, ansatz depth, entangling topology, number of retained latent dimensions) would allow more efficient exploration of the design space than manual or grid search, particularly given the high per-run cost of circuit simulation.
- **Rigorous, shift-aware, multi-institutional evaluation.** Extending evaluation to larger, multi-site datasets (e.g., CheXpert, NIH ChestX-ray14) and explicit domain-adaptation techniques (reweighting, adversarial domain adaptation) would provide a more realistic assessment of generalization than single-institution pediatric data [39].
- **Ensemble and hybrid-decision strategies.** Combining classical and quantum model outputs (e.g., via weighted logit averaging or a meta-classifier) could exploit differences in the error profiles of the two approaches, potentially improving both sensitivity and specificity relative to either model alone [16].
- **Quantum-side interpretability.** Developing sensitivity- and perturbation-based tools analogous to Grad-CAM for the quantum decision layer would help clarify whether the VQC learns complementary or redundant information relative to the classical autoencoder projection, and would support the broader goal of building trustworthy clinical decision-support systems.

10 Conclusion

This paper reviewed the computational and algorithmic foundations of hybrid quantum–classical architectures for pneumonia detection from chest X-ray images, with an emphasis on data-encoding strategies, variational and convolutional ansatz design, quantum kernel methods, and the trainability limitations imposed by barren plateaus together with their principal mitigation strategies. We surveyed a growing body of literature applying hybrid QML pipelines to CXR tasks—including recent works demonstrating statistically significant improvements over classical baselines [14] and near-unity accuracy on COVID-19 radiography datasets [30]—while noting the evaluation limitations that temper direct comparison across studies.

Using a ConvNeXt-Tiny/autoencoder/VQC pipeline evaluated on a public pediatric CXR dataset as a concrete case study, we showed that a 6-qubit, 62-parameter variational quantum classifier achieves a level of diagnostic performance comparable to—though not superior to—a classical MLP of comparable input dimensionality. On a patient-disjoint held-out test set of 624 images under an approximately 11-percentage-point class-prevalence shift, the VQC attained a balanced accuracy of 0.747 against 0.729 and an accuracy of 74.84% against 74.20%, with approximately $34\times$ fewer trainable parameters in the decision layer (62 versus 2,113), while the classical baseline attained the higher threshold-independent ranking quality (test AUC-ROC 0.848 against 0.806); an exact McNemar test found the accuracy difference insignificant ($p = 0.71$). This result should be read narrowly: it demonstrates that a compact variational circuit can serve as a functional, parameter-efficient classifier on a compressed medical-imaging representation, not that a computational or statistical quantum advantage has been achieved. Barren plateaus, state-preparation overhead, aggressive dimensionality-reduction-induced information loss, and dataset shift all constrain current performance.

The hardware execution of Section 6 sharpens this conclusion into a negative result. On the physical processor tested, noise compressed the span of the circuit’s output by 59% relative to

the noise-free simulator, and zero-noise extrapolation restored that span without restoring per-sample predictions. The gate accounting explains why: 77.5% of the circuit’s two-qubit gates belong to an amplitude-encoding routine whose depth grows exponentially in the register and which no degree of ansatz parameter-efficiency can offset. For hybrid pipelines of this shape the binding constraint on present hardware is state preparation, not classifier capacity—and simulation-based studies, including the simulator results reported here, systematically understate it by treating state preparation as free.

We regard hybrid VQC decision layers, quantum convolutional architectures, and quantum kernel classifiers as promising but early-stage research directions whose eventual clinical or practical relevance will depend on: progress in fault-tolerant or higher-fidelity NISQ hardware; encodings whose depth does not grow exponentially in the register; more expressive yet trainable ansatz designs (e.g., data re-uploading, low-entanglement initialization); hardware-aware error mitigation validated at the level of individual predictions rather than distribution summaries; and substantially more rigorous, multi-institutional, shift-aware evaluation protocols than are currently standard in the literature.

References

- [1] Aaronson, S. (2015). Read the fine print. *Nature Physics*, 11(4), 291–293.
- [2] Biamonte, J., Wittek, P., Pancotti, N., Rebentrost, P., Wiebe, N., & Lloyd, S. (2017). Quantum machine learning. *Nature*, 549(7671), 195–202.
- [3] Centers for Disease Control and Prevention. (2025). *About Pneumonia*. <https://www.cdc.gov/pneumonia/about/index.html>
- [4] Cerezo, M., Sone, A., Volkoff, T., Cincio, L., & Coles, P. J. (2021). Cost-function-dependent barren plateaus in shallow parametrized quantum circuits. *Nature Communications*, 12(1), 1791.
- [5] Cong, I., Choi, S., & Lukin, M. D. (2019). Quantum convolutional neural networks. *Nature Physics*, 15(12), 1273–1278.
- [6] Decoodt, P., Liang, T. J., et al. (2023). Hybrid classical–quantum transfer learning for cardiomegaly detection in chest X-rays. *Journal of Imaging*, 9(7), 128. <https://doi.org/10.3390/jimaging9070128>
- [7] Ding, Y., Zhang, S., & Li, X. (2024). Noise mitigation: an efficient approach for quantum convolutional neural networks to enhance performance in NISQ. Preprint. <https://doi.org/10.21203/rs.3.rs-4594670/v1>
- [8] Haug, T., Self, C. N., & Kim, M. (2021). Quantum machine learning of large datasets using randomized measurements. *arXiv preprint arXiv:2108.01039*.
- [9] Havlíček, V., Córcoles, A. D., Temme, K., Harrow, A. W., Kandala, A., Chow, J. M., & Gambetta, J. M. (2019). Supervised learning with quantum-enhanced feature spaces. *Nature*, 567(7747), 209–212.
- [10] He, K., Zhang, X., Ren, S., & Sun, J. (2016). Deep residual learning for image recognition. In *Proceedings of the IEEE Conference on Computer Vision and Pattern Recognition (CVPR)*, 770–778.
- [11] Henderson, M., Shakya, S., Pradhan, S., & Cook, T. (2020). Quantum convolutional neural networks: Powering image recognition with quantum circuits. *Quantum Machine Intelligence*, 2(1), 2.

- [12] Institute for Quality and Efficiency in Health Care (IQWiG). (2021). *Overview: Pneumonia*. InformedHealth.org. <https://www.ncbi.nlm.nih.gov/books/NBK525774/>
- [13] Kashif, M., & Al-Kuwari, S. (2023). The unified effect of data encoding, ansatz expressibility and entanglement on the trainability of HQNNs. *International Journal of Parallel, Emergent and Distributed Systems*, 38(5), 362–400. <https://doi.org/10.1080/17445760.2023.2231163>
- [14] Kulkarni, V., Pawale, S., & Kharat, A. (2023). A classical–quantum convolutional neural network for detecting pneumonia from chest radiographs. *Neural Computing and Applications*, 35(21), 15503–15510. <https://doi.org/10.1007/s00521-023-08566-1>
- [15] Khatoniar, R., Vyas, V., Acharya, V., et al. (2024). Quantum convolutional neural network for medical image classification. In *Proceedings of the 2nd International Conference on Trends in Quantum Computing and Emerging Business Technologies (TQCEBT)*. IEEE. <https://doi.org/10.1109/tqcebt59414.2024.10545121>
- [16] Kundu, R., Das, R., Geem, Z. W., Han, G.-T., & Sarkar, R. (2021). Pneumonia detection in chest X-ray images using an ensemble of deep learning models. *PLOS ONE*, 16(9), e0256630.
- [17] Liu, X., Liu, G., Huang, J., et al. (2022). Mitigating barren plateaus of variational quantum eigensolvers. *arXiv preprint arXiv:2205.13539*.
- [18] Ortiz Marrero, C., Kieferová, M., & Wiebe, N. (2021). Entanglement-induced barren plateaus. *PRX Quantum*, 2(4), 040316. <https://doi.org/10.1103/PRXQuantum.2.040316>
- [19] McClean, J. R., Boixo, S., Smelyanskiy, V. N., Babbush, R., & Neven, H. (2018). Barren plateaus in quantum neural network training landscapes. *Nature Communications*, 9(1), 4812.
- [20] Proctor, B., Cai, Y., Houghton, K., Kumar, A., Dahl, E., & Johnson, P. (2021). Towards the quantum verification of quantum algorithms. *Nature Physics*, 17(6), 542–547. <https://doi.org/10.1038/s41567-020-0930-1>
- [21] Mitarai, K., Negoro, M., Kitagawa, M., & Fujii, K. (2018). Quantum circuit learning. *Physical Review A*, 98(3), 032309.
- [22] Möttönen, M., Vartiainen, J. J., Bergholm, V., & Salomaa, M. M. (2005). Transformation of quantum states using uniformly controlled rotations. *Quantum Information and Computation*, 5(6), 467–473. <https://arxiv.org/abs/quant-ph/0407010>
- [23] Naz, A. (2020). Literature review on X-ray based pneumonia detection using deep learning. *LCJSTEM*, 1(1), 1–9.
- [24] Nielsen, M. A., & Chuang, I. L. (2010). *Quantum Computation and Quantum Information* (10th anniversary ed.). Cambridge University Press.
- [25] Patti, T. L., Najafi, K., Gao, X., & Yelin, S. F. (2021). Entanglement devised barren plateau mitigation. *Physical Review Research*, 3, 033090. <https://doi.org/10.1103/PhysRevResearch.3.033090>
- [26] Pérez-Salinas, A., Cervera-Lierta, A., Gil-Fuster, E., & Latorre, J. I. (2020). Data re-uploading for a universal quantum classifier. *Quantum*, 4, 226.

- [27] Pesah, A., Cerezo, M., Wang, S., Volkoff, T., Sornborger, A. T., & Coles, P. J. (2021). Absence of barren plateaus in quantum convolutional neural networks. *Physical Review X*, 11(4), 041011.
- [28] Preskill, J. (2018). Quantum computing in the NISQ era and beyond. *Quantum*, 2, 79.
- [29] Beaulieu, D., Miracle, D., Pham, A. T., & Scherr, W. (2022). Quantum kernel for image classification of real-world manufacturing defects. *arXiv preprint arXiv:2212.08693*.
- [30] Rao, G. V. E., Rajitha, B., Srinivasu, P. N., Ijaz, M. F., & Woźniak, M. (2024). Hybrid framework for respiratory lung diseases detection based on classical CNN and quantum classifiers from chest X-rays. *Biomedical Signal Processing and Control*, 88, 105567. <https://doi.org/10.1016/j.bspc.2023.105567>
- [31] Schuld, M., & Petruccione, F. (2018). *Supervised Learning with Quantum Computers*. Springer, Quantum Science and Technology.
- [32] Schuld, M., & Killoran, N. (2019). Quantum machine learning in feature Hilbert spaces. *Physical Review Letters*, 122(4), 040504.
- [33] Schuld, M. (2021). Supervised quantum machine learning models are kernel methods. *arXiv preprint arXiv:2101.11020*.
- [34] Ortiz-Toro, C., García-Pedrero, A., Lillo-Saavedra, M., & Gonzalo-Martín, C. (2022). Automatic detection of pneumonia in chest X-ray images using textural features. *Computers in Biology and Medicine*, 145, 105466. <https://doi.org/10.1016/j.combiomed.2022.105466>
- [35] Sharma, A., et al. (2023). Survey of encoding techniques for quantum machine learning. *Cybernetics and Physics*, 12(1), 45–67.
- [36] Sharma, R. (2023). Classification of pneumonia via a hybrid ZFNet–quantum neural network. *Current Medical Imaging Reviews*. Advance online publication.
- [37] Sim, S., Johnson, P. D., & Aspuru-Guzik, A. (2019). Expressibility and entangling capability of parametrized quantum circuits for hybrid quantum-classical algorithms. *Advanced Quantum Technologies*, 2(12), 1900070.
- [38] Wei, S., Chen, Y., Zhou, Z.-R., & Long, G.-L. (2022). A quantum convolutional neural network on NISQ devices. *AAPPS Bulletin*, 32(1), 2. <https://doi.org/10.1007/s43673-021-00030-3>
- [39] Zech, J. R., Badgeley, M. A., Liu, M., Costa, A. B., Titano, J. J., & Oermann, E. K. (2018). Variable generalization performance of a deep learning model to detect pneumonia in chest radiographs: A cross-sectional study. *PLOS Medicine*, 15(11), e1002683.
- [40] Zhu, Y., Liang, Y., & Situ, H. (2024). Enhancing variational quantum circuit training: an improved neural network approach for barren plateau mitigation. *arXiv preprint arXiv:2411.09226*.